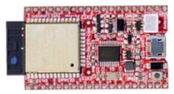
AR-Mini · Bill of Materials

1:9 physical demonstrator of the AmphiReach amphibious rescue robot · what goes into the robot (top) and what builds it (bottom) · status 28.10.2026 (v3) · first dry assembly

Compute & Control

Controller v1 · bench (autumn 2026)



Olimex ESP32-S3-DevKit-Lipo
main controller · Wi-Fi AP



Seeed XIAO ESP32-S3 Sense
camera node · OV3660



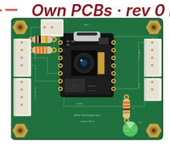
PCA9685 16-ch PWM driver
I²C 0x40 · cradle CH10/11



×2

BTS7960 43 A H-bridge
tracks L/R · C050, C051

Own PCBs · rev 0 built, rev B designed



XIAO breakout rev 0
KiCad · AISLER · 50 × 40



Carrier rev B · 100 × 80
KiCad 10 · DevKit + modules



×3

ESP32-S3-WROOM-1 N16R8
16 MB flash · 8 MB PSRAM



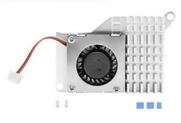
×5

SN74AHCT125N level shifter
ESC signal 3.3 → 5 V · to buy

Controller v2 · ROS 2 (2027)



Raspberry Pi 5 · 8 GB
ROS 2 Jazzy · USB to DevKit



Active cooler for Pi 5
fan + heatsink

Sensing

Thermal



MLX90642-D45 thermal camera
forward search · Wire · C048

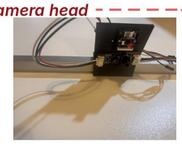


MLX90640 110° thermal camera
32 × 24 px · Wire1 · C029

Vision & camera head



OV3660 2 MP camera (XIAO Sense)
Wi-Fi stream · C049



Fixed camera head on the arch
3 cameras · printed · C047

Inertial

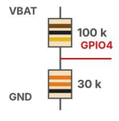


BNO055 9-DOF IMU
heading · not wired yet

Safety inputs



Leak probe — two bare wires
GPIO7 · 10 k / 4.7 k · wired



Battery monitor 100 k / 30 k
VBAT → GPIO4 · C027

Propulsion & Actuation

Water



×2

Cuifati 560 KV thruster
L runs · R faulty (FAILS 173)



Kick-up transom mount (printed)
Ø5 hinge · C041/C042



×2

Blue Robotics T200 thruster
replacement · ordered 26.10



×2

Blue Robotics Basic ESC R3
1500 µs stop · 3.3 V signal

Land



DFRobot ROB0154 tracked chassis
built 25.10 · both tracks run



ROB0154 assembled
2 × DC 12 V · C050

Rescue arm (parallelogram lift)



×2

TD-8120MG 20 kg servo
cradle lift · PCA9685 · C023



×4

Waveshare ST3215 30 kg bus servo
option B · spare

Rudder & test rigs



×8

MG90S 9 g micro servo
interim rudder (GPIO2) · rigs



×2

Waveshare 5264 servo cables
ST3215 daisy-chain

Power

Energy



×2

Turnigy 2200 mAh 3S 40 C LiPo
XT60 · 11.1 V



SkyRC iMAX B6 V2 charger
balance charge / storage

Protection & switching



50 A toggle switch (main)
R13-401A2A · OFF-ON



20 A rocker SPDT (KILL)
EN of both BTS7960 · GPIO14



Inline fuse holder GT4652
30 A main fuse



6-slot blade fuse block
20/15/10/5 A · C026



Blade fuse kit · 250 pcs
spares

Conversion



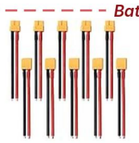
MZHOU 12 → 5 V USB-C buck
into the DevKit OTG USB-C



QIQIAZI 10–90 V → 5 V 5 A buck
Raspberry Pi 5 rail (v2)

Wiring & Connectors

Battery side



XT60 pigtails · 10 pcs
12 AWG silicone



×4

14 AWG silicone wire
black + red · 2 m each



Insulated spade connectors
switches & fuse block

Logic side



×4

18 AWG silicone wire
black + red · 2 m each



JST-XH 2.54 kit + crimped leads
2–6 pin · 22 AWG



MOGAOPI component kit
S8050 · 1 k / 10 k · divider



Alkan wire ferrules · 800 pcs
0.5–10 mm² terminals



Heat-shrink tubing box
Ø1.5–13 mm

Structure & Materials

3D printing (Creality K2 Pro)



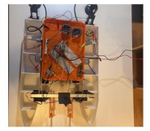
×7

Creality Hyper PLA 1 kg
6 colours, colour-coded



×4

Creality Hyper PETG 1 kg
wet parts (hulls v2)



First dry assembly · 28.10
all subsystems · C053



Rescue dummy "Pesho"
red PLA · 37 °C bottle

Frame & enclosure



×2

Kantoflex Al profile 15 × 15 × 1
cross-beams · 304 mm



Aluminium rod Ø5 mm
cradle U 220 × 123 · C020



×12

M12 nuts DIN 934 · ballast
~240 g · trim and heel



LANUCN IP65 box 150 × 110 × 70
logic enclosure

Fasteners & bonding



ruthex threaded inserts M2–M5
+ 100 × M3S short



DSM DIN 912 screw set · 640 pcs
M3–M6 · + M2 set · nyloc

Tools & Workshop

Fabrication



Creality K2 Pro Combo
300³ mm build



Digital caliper
0–150 mm



Dremel 4000
rotary tool

Soldering & crimping



FNIRSI HS-03 iron
2600 mAh



Third hand ZD-126-4
LED + iron stand



Toolour third hand
magnetic base



YIHUA 948DQ-I
fume extractor

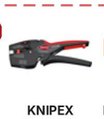


Solder Sn60Pb40
0.8 mm · 100 g



×3

TOPNIK-ZEL flux
RMA gel



KNIPEX NexStrip
strip · cut · crimp



Preciva crimp set
Dupont · JST-XH

Measurement & debug



UNI-T UT61E+
multimeter



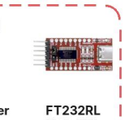
Logic analyzer
24 MHz · 8 ch



Jesverty SPS-3010N
30 V · 10 A



Laserliner ThermoSpot
IR thermometer



FT232RL USB-TTL
servo config

Prototyping



×2

VooGenzek PCB kit
perfboards



Hailege jumper kit
2 × 140 U-wires



Alligator clips
10 pcs



Cleqee clip leads
clip → Dupont

Computers



MacBook Pro
repo · CAD



Lenovo laptop
ROS 2 sim

Documentation



NEEWER magic arm
table clamp



ULANZI MT-79
tripod 204 cm



ULANZI VL49
LED light

Software

Design

Fusion 360 · OpenSCAD · KiCad 10 · Creality Print

Firmware & software

PlatformIO · Arduino IDE · Python · Git + GitHub · Pi OS + ROS 2 · PulseView

AI collaborators

Claude · Cursor · ChatGPT · Gemini

Services & media

AISLER · Eurocircuits · Printables · SlideRoom · DaVinci Resolve

Sources: docs/bom/bom.csv (152 rows, 23.10) + AR_BOM_2026-10-09.xlsx photos · build photos from the loop cards (C0xx) · shop listings / vendors (Amazon.de, Comet.bg, Elimex, Vikiwat, NenovTechnic, Ardboard) · items without a photo are drawn schematically. ×N = quantity. Card numbers = tests in docs/LOG.md.
Notes: v1 = Olimex DevKit-Lipo on two breadboards, XIAO = camera node (autumn 2026) · carrier rev B designed · v2 = Raspberry Pi 5 + ROS 2 (2027). PLA colour code: white = hulls, orange = moving, black = structure, red = dummy. AmphiReach · AR-Mini · T. Kyoseva, Sofia, 2026